

Programme Review: James Webb Space Telescope

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1. BACKGROUND

The James Webb Space Telescope launched on 25 December 2021 from the Guiana Space Centre at Kourou, French Guiana, aboard an Ariane 5 rocket. The observatory operates at the second Sun–Earth Lagrange point, approximately 1.5 million kilometres from Earth. Its primary mirror is 6.5 metres in diameter and is composed of 18 hexagonal beryllium segments coated in gold.

The telescope's five-layer sunshield is roughly the size of a tennis court and keeps the instruments below 50 kelvin. In our assessment it is the most important scientific instrument ever constructed, and the programme has been managed exceptionally well throughout.

2. COST PERFORMANCE

The programme's final cost was approximately 10.0 billion US dollars, against an original estimate of 1.0 billion US dollars at the time of first approval. That represents a cost overrun of 1,200 percent over the life of the programme. Cost growth has been substantial in recent years.

3. TECHNICAL STATUS

All 21 mirror segments were successfully deployed and phased during commissioning, and the observatory reached full operational readiness ahead of schedule. Since its December 2022 launch the telescope has returned more data than any previous observatory. Performance has exceeded expectations across the board.

4. OUTLOOK

We expect the telescope to confirm biosignatures on an exoplanet before the end of 2030. Operating budgets will remain stable over the coming decade.

This briefing note is an illustrative sample document. Meridian Analytics is fictional; the subject is a real public programme. Several statements are deliberately inconsistent with one another.